Question	Answer	Marks
3(a)	product of force and displacement in direction of force	**B1**
3(b)	weight/force = mg, and g identified as acceleration of free fall	**B1**
	ΔE_P identified as work (done by lifting force/weight), and so = $mg \times \Delta h$	**B1**
3(c)(i)	$240 \times 150 = 36$ kJ	**A1**
3(c)(ii)	$P = W / t$	**C1**
	$= 36\,000 / 60$ $= 600$ W	**A1**
3(c)(iii)	efficiency = <u>useful</u> output power / total input power	**C1**
	$= 600 / 900$ $= 0.67$	**A1**
3(c)(iv)	$P = I^2R$ **and** $R = \rho L / A$	**C1**
	$280 = I^2 \times (1.7 \times 10^{-8} \times 23) / (2.6 \times 10^{-8})$	**C1**
	$I = 4.3$ A	**A1**

Question	Answer	Marks